

system optimized specifically to store virtual machine images and ensure high I/O throughput. This allows for extreme levels of performance. For example, VMware demonstrated back in 2011 that a single ESX Server could issue 1 million disk operations per second (VMware, 2011).

5. ESX Server made it easy to introduce new capabilities, which required the tight coordination and specific configuration of multiple components of a computer. For example, ESX Server introduced VMotion, the first virtualization solution that could migrate a live virtual machine from one machine running ESX Server to another machine running ESX Server, while it was running. This achievement required the coordination of the memory manager, the CPU scheduler, and the networking stack.

Over the years, new features were added to ESX Server. ESX Server evolved into ESXi, a small-footprint alternative that is sufficiently small in size to be preinstalled in the firmware of servers. Today, ESXi is VMware's most important product and serves as the foundation of the vSphere suite.

7.12 RESEARCH ON VIRTUALIZATION AND THE CLOUD

Virtualization technology and cloud computing are both extremely active research areas. The research produced in these fields is way too much to enumerate. Each has multiple research conferences. For instance, the Virtual Execution Environments (VEE) conference focuses on virtualization in the broadest sense. You will find papers on migration deduplication, scaling out, and so on. Likewise, the ACM Symposium on Cloud Computing (SOCC) is one of the best-known venues on cloud computing. Papers in SOCC include work on fault resilience, scheduling of data center workloads, management and debugging in clouds.

Hardware support for virtualization is now present in nearly all relevant CPU architectures, restoring the Popek and Goldberg architectural principles into practice. Notably, ARM added a new privilege level "EL2" to support hardware virtualization in ARMv8. In mobile platforms, virtualization is often deployed with another hardware feature called TrustZone to allow a "secure" virtual machine to co-exist with the main operating environment (Dall et al., 2016).

Security is perpetually a hot topic for research (Dai et al., 2020; Trach et al., 2020), as is reducing energy usage (Kaffes et al., 2020). With so many data centers now using virtualization technology, the networks connecting these machines are also a major subject of research (Alvarez et al., 2020).

One of the nice things about virtualization hardware is that untrusted code can get direct but safe access to hardware features like page tables, and tagged TLBs. With this in mind, the Dune project (Belay, 2012) did not aim to provide a machine abstraction, but rather a process abstraction. The process is able to enter Dune

mode, an irreversible transition that gives it access to the low-level hardware. Nevertheless, it is still a process and able to talk to and rely on the kernel. The only difference that it uses the `VMCALL` instruction to make a system call. The Dune approach was later used for research in dataplane operating systems such as IX (Belay, 2017) and finally adapted as the foundation for Google’s container solution gVisor (Young, 2019).

Speaking of containers, the cloud is seeing a shift from being a platform for tenants to run virtual machines (specified by a virtual disk image) to a platform used by tenants to run containers specified as Dockerfiles and coordinated by orchestrators such as Kubernetes. Often, the container runs within a virtual machine but the guest operating system is now run by the cloud provider. The latest trend, called “serverless”, further decouples the application logic from its environment (Shahrad et al., 2020). The idea here is to allow a service to be automatically spun up to serve a single function (an RPC over HTTPS) without managing any aspect of its operating system environment. Amazon calls its serverless technology firecracker (Barr, 2018) and Google calls it gVisor (Young, 2019). Serverless computing has gained even more popularity with the adoption of the Function-as-a-Service (FAAS) model (Kim and Lee, 2019). Of course, the operating system and indeed the server both still exist in serverless operations. But they are hidden from the developer.

The cloud is therefore much more complex than it was in the beginning with the original AWS EC2 model of virtual machines: networks are virtualized, and applications are encapsulated into containers and serverless models that decouple them from the underlying layers. The cloud is more complex, but also more powerful and central to nearly every computing organization. With this importance comes another consideration: trust. Specifically, how much must a tenant trust the cloud service provider? The correct answer is obviously “as little as necessary.”

To achieve this goal of minimal trust dependencies, Intel introduced SGX as the first architectural extension for “confidential computing.” SGX creates enclaves, which are isolated in hardware from the host operating system and other applications, with the content of memory and registers cryptographically encrypted by the hardware. In a way, technologies such as SGX are similar to virtualization extensions: they create new ways for software to isolate itself from each other. SGX has notably been used in research to run entire operating systems such as in Haven (Baumann, 2015) or Linux containers such as in SCONE (Arnautov, 2016).

7.13 SUMMARY

Virtualization is the technique of simulating a computer, but with high performance. Typically one computer runs many virtual machines at the same time. This technique is widely used in data centers to provide cloud computing. In this chapter we looked at how virtualization works, especially for paging, I/O, and multicore systems. We also studied an example: VMWare.